

Transformer Fleet Health & Analytics Report

Asset Condition Assessment & Technical Interview Guide | Sample Size: 500 Substation Transformers (11.0 kV)

TOTAL MONITORED

500

HEALTHY STATE

155 (31.0%)

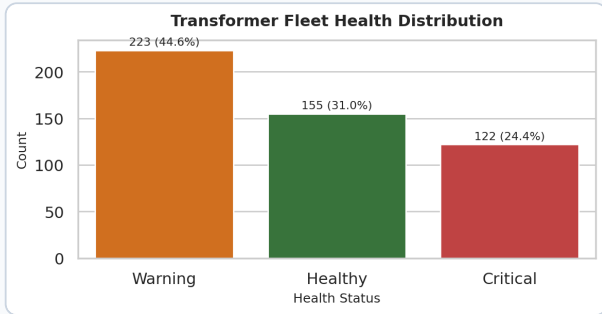
WARNING STATE

223 (44.6%)

CRITICAL ACTION

122 (24.4%)

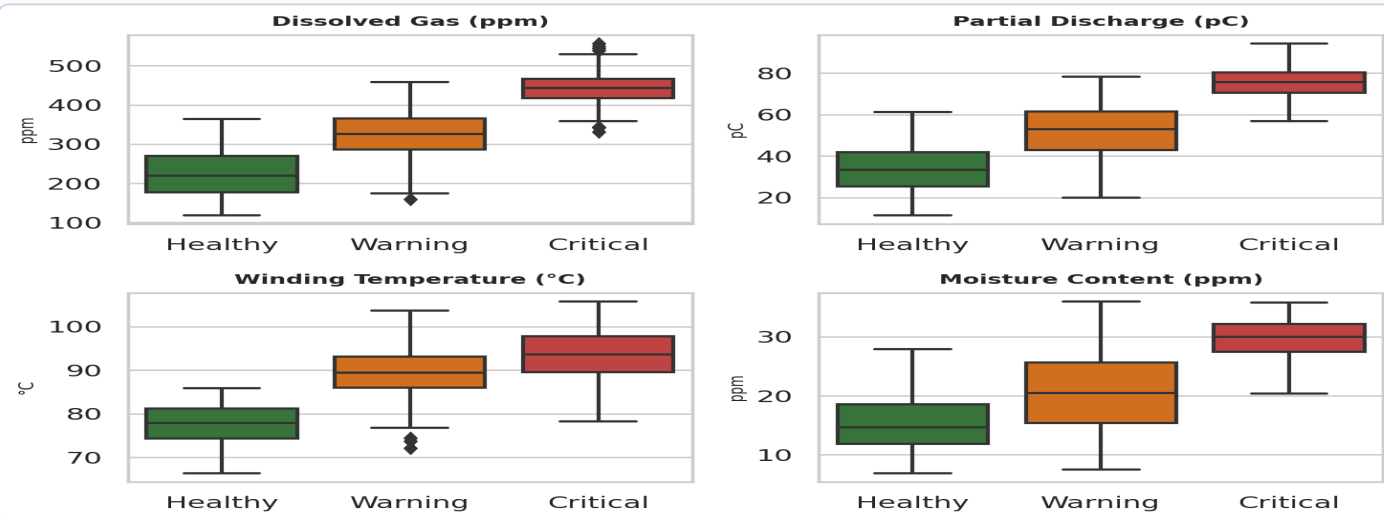
1. Fleet Distribution



2. Key Parameter Averages

Parameter	Healthy	Warning	Critical
Mean Age (Years)	8.3 yrs	14.0 yrs	25.4 yrs
Operating Load (%)	51.6%	73.4%	75.6%
Winding Temp (°C)	77.8°C	89.4°C	93.6°C
Dissolved Gas (ppm)	223.4 ppm	325.1 ppm	445.5 ppm
Moisture (ppm)	15.4 ppm	20.5 ppm	29.5 ppm
Partial Discharge (pC)	33.5 pC	51.8 pC	75.7 pC
Vibration (mm/s)	2.39 mm/s	3.19 mm/s	4.20 mm/s

3. Diagnostic Key Indicators Distribution



Key Engineering Insight: Aging units (>20 years) under continuous high loading (>75%) show severe insulation degradation. High partial discharge (>70 pC) combined with dissolved gas spikes (>400 ppm) indicates immediate thermal breakdown risk of paper-oil insulation.

4. Technical & Electrical Interview Questions

Q1: Which parameters are the strongest predictors of critical transformer failure?

Answer: Dissolved Gas (DGA) ($r = 0.90$), Partial Discharge ($r = 0.93$), and Moisture Content ($r = 0.97$) show the highest correlation with health degradation and asset age (25.4 years average for critical units).

Q2: Why does moisture ingress accelerate transformer insulation breakdown?

Answer: Moisture reduces the dielectric strength of transformer oil and accelerates thermal depolymerization of kraft paper insulation, leading to internal short circuits and reduced lifetime.

Q3: How would you handle feature selection for machine learning using this dataset?

Answer: Drop `Vo1tage_kV` as it remains constant (11.0 kV, variance = 0). Use multi-collinearity filtering on highly correlated thermal pairs (Oil Temp vs. Winding Temp) and select tree-based algorithms (XGBoost/Random Forest).

Q4: What immediate maintenance strategy would you recommend to the utility grid team?

Answer: Prioritize the 122 Critical units (24.4%) for online oil filtration and DGA gas chromatography testing. Re-balance grid loads on units exceeding 80% load to reduce winding temperatures below 85°C.